

《语言作为世界的接口》
《Language as the World Interface》

Language as the Interface to the World
/ˈlæŋɡwɪdʒ æz ði ˈɪntəfeɪs tuː ðə wɜːld/

Language as the Interface of the World
/ˈlæŋɡwɪdʒ æz ði ˈɪntəfeɪs əv ðə wɜːld/

Language as the World Interface
/ˈlæŋɡwɪdʒ æz ðə wɜːld ˈɪntəfeɪs/

Language as an Interface for World Modelling
/ˈlæŋɡwɪdʒ æz ən ˈɪntəfeɪs fə wɜːld ˈmɒdəlɪŋ/

Language as the Mediating Interface of Reality
/ˈlæŋɡwɪdʒ æz ðə ˈmiːdiətɪŋ ˈɪntəfeɪs əv rɪˈælɪti/

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Introduction: Why Start with “Interface” Instead of “Consciousness”

导论：为什么从“接口”而不是“意识”谈起

1. Consciousness Is a Bad Entry Point

1. 意识不是一个好的切入点

English

Consciousness is often treated as the starting point when discussing the relationship between humans and the world.

From a modelling perspective, this choice immediately creates problems.

中文

在讨论人类与世界关系时，“意识”常被当作起点。
但从建模角度看，这个选择一开始就制造了问题。

1.1 Consciousness Lacks a Clear Boundary

1.1 意识缺乏清晰边界

English

A system can only be modelled if its boundary can be specified.
Consciousness does not provide such a boundary.

Where does consciousness begin?
Where does it end?
Which processes are inside, and which are outside?

These questions have no operational answers.

中文

一个系统只有在边界可定义时，才能被建模。
而意识并不提供这样的边界。

意识从哪里开始？
到哪里结束？
哪些过程在“意识之内”，哪些在外？

这些问题没有可操作的答案。

1.2 Consciousness Is Not Alignable

1.2 意识不可对齐

English

Conscious experience is private by definition.
There is no shared reference frame that allows two systems to align their consciousness directly.

Without alignment, comparison collapses.
Without comparison, modelling collapses.

中文

意识体验在定义上就是私有的。
不存在一个共享参考系，可以让两个系统直接对齐各自的意识。

无法对齐，就无法比较；
无法比较，就无法建模。

1.3 Consciousness Is Not Directly Operable

1.3 意识不可直接操作

English

A system cannot issue commands to “consciousness itself”.
It can only operate on what consciousness exposes.

This distinction already implies an intermediate layer.

中文

系统无法直接对“意识本身”下指令。
它只能操作意识所暴露出来的内容。

这一点本身就暗示了：
存在一个中间层。

2. Why “Interface” Is a Better Starting Concept

2. 为什么“接口”是更好的起点

2.1 Interface Is a First-Class System Concept

2.1 接口是系统论中的一级概念

English

In systems theory, an interface is not an optional detail.
It is a structural necessity.

An interface defines:

what can be exchanged,

in what form,

under what constraints.

中文

在系统论中，接口不是可选细节。
它是结构上的必需品。

接口定义了：

什么信息可以被交换，

以什么形式交换，

在什么约束条件下交换。

2.2 Interface Exists Before Understanding

2.2 接口先于理解存在

English

A system can use an interface without understanding how it works internally.

You can speak without knowing acoustics.

You can think without knowing neurochemistry.

You can communicate without accessing another system's consciousness.

中文

系统可以在不理解接口内部机制的情况下使用接口。

你可以说话，却不了解声学；

你可以思考，却不了解神经化学；

你可以交流，却无法进入他人的意识。

2.3 Black-Box Interaction Is the Norm

2.3 黑箱交互才是常态

English

Most real-world interactions are black-box interactions.
Interfaces make this possible by hiding complexity.

Understanding is optional.
Connectivity is not.

中文

现实世界中的大多数交互都是黑箱交互。
接口通过屏蔽复杂性，使这种交互成为可能。

理解是可选的，
接得上不是。

3. Minimal Requirements for World–System Interaction

3. 世界—系统交互的最小要求

Before identifying a candidate interface, we need criteria.

3.1 Discreteness

3.1 离散性

English

The unit must be discretisable.
Without discreteness, repetition and reuse are impossible.

中文

交互单位必须可离散化。
否则无法重复，也无法复用。

3.2 Stability Across Time

3.2 跨时间稳定性

English

The unit must survive beyond the immediate moment.
Ephemeral signals cannot support accumulated interaction.

中文

交互单位必须能跨越瞬时存在。
瞬态信号无法支撑长期交互。

3.3 Shareability Across Systems

3.3 跨系统可共享性

English

A world interface must allow multiple systems to align on it.
Purely private units fail this requirement.

中文

世界接口必须允许多个系统对齐。
纯私有的单位无法满足这一点。

3.4 Compositionality

3.4 组合性

English

The unit must combine with others to form higher-order structures.
Otherwise, complexity explodes.

中文

交互单位必须具备组合能力。
否则复杂性会指数爆炸。

4. What This Introduction Does Not Claim

4. 本导论没有主张什么

English

This introduction does not claim:

that consciousness is unreal,

that language is the only cognitive process,

or that all experience is linguistic.

It only claims a narrower point.

中文

本导论并未主张：

意识是虚假的，

语言是唯一的认知过程，

一切经验都是语言性的。

它只提出一个更窄的判断。

5. Working Hypothesis

5. 工作假设

English

There exists a layer between system and world that:

is not raw perception,

is not direct action,

yet enables stable, shared, and compositional interaction.

The rest of this work tests whether language occupies that layer.

中文

在系统与世界之间，存在这样一层结构：

不是原始感知，

不是直接行为，

却能够实现稳定、共享、可组合的交互。

接下来的内容，只做一件事：

检验语言是否占据了这一层。

Chapter 1: World, System, and Interface — Basic Definitions

第一章：世界、系统与接口的基本定义

1. What Is the “World”

1. 什么是“世界”

1.1 The World Is Not “Everything That Exists”

1.1 世界不是“一切存在之物”

English

In everyday language, “the world” often means all of reality.

For modelling purposes, this definition is useless.

A system never interacts with total reality.

It only interacts with what lies outside its boundary.

中文

在日常语言中，“世界”常被理解为全部现实。

但在建模中，这一定义是不可用的。

系统从不与“全部现实”交互。

它只与自身边界之外的部分交互。

1.2 The World Is Defined Relative to a System

1.2 世界是相对于系统定义的

English

There is no system-independent “world” in operational terms.

The world is whatever a system cannot directly access or control.

Change the system, and the world changes.

中文

在可操作意义上，不存在与系统无关的“世界”。

世界指的是系统无法直接访问或控制的部分。

系统一变，世界也随之改变。

1.3 The World Is Always Larger Than Access

1.3 世界永远大于可访问范围

English

No system has full access to its world.

Incomplete access is not a limitation—it is a structural fact.

Uncertainty is not a bug.

It is a consequence of bounded systems.

中文

没有任何系统可以完全访问其世界。

这种不完全性不是缺陷，而是结构事实。

不确定性不是错误，

而是有边界系统的必然结果。

1.4 “Unknown World” Is Not a Philosophical Problem

1.4 “未知世界”不是哲学问题

English

The existence of unknown aspects of the world does not require metaphysical explanation.

It follows directly from boundary conditions.

中文

世界中存在未知部分，并不需要形而上学解释。

它直接源自系统的边界条件。

2. What Is a “System”

2. 什么是“系统”

2.1 A System Is Defined by Its Boundary

2.1 系统由其边界定义

English

A system exists if and only if a boundary can be specified.

The boundary determines:

what is internal,

what is external,

and what can be exchanged.

中文

只有当边界可被定义时，系统才存在。

边界决定了：

什么是内部，

什么是外部，

以及什么可以被交换。

2.2 A System Is Not Necessarily a Physical Object

2.2 系统不一定是物理实体

English

A system can be biological, mechanical, social, or abstract.
Its physical form is secondary to its functional closure.

中文

系统可以是生物的、机械的、社会的，甚至是抽象的。
其物理形态并不重要，功能闭合才是关键。

2.3 Internal State Is Not the World

2.3 内部状态不是世界

English

A system maintains internal states to cope with the world.
These states are representations, not the world itself.

Confusing internal state with the world is a category error.

中文

系统通过内部状态来应对世界。
这些状态是表示，而不是世界本身。

将内部状态当作世界，是范畴错误。

2.4 The System–World Divide Is Non-Negotiable

2.4 系统—世界划分不可取消

English

Any attempt to dissolve the system–world boundary eliminates the system itself.

No boundary → no system → no interaction.

中文

任何试图抹消系统—世界边界的做法，
都会同时抹消系统本身。

无边界 → 无系统 → 无交互。

3. Why the System Cannot Touch the World Directly

3. 为什么系统无法直接接触世界

3.1 Direct Access Would Require Boundary Collapse

3.1 直接访问意味着边界坍塌

English

If a system could access the world directly,
there would be no distinction between internal and external.

The system would cease to be a system.

中文

如果系统能够直接访问世界，
内部与外部的区分将不复存在。

系统本身也将不复存在。

3.2 Interaction Requires Mediation

3.2 交互必然需要中介

English

Because direct access is impossible, all interaction must be mediated.
This mediation layer is not optional.

中文

由于直接访问不可能，
所有交互都必须经过中介层。

这一中介层不是可选项。

3.3 What an “Interface” Is

3.3 什么是“接口”

3.3.1 An Interface Is Not a Channel

3.3.1 接口不是通道

English

A common misunderstanding treats an interface as a passive channel through which information flows.

This view is incorrect.

A channel only transmits.

An interface constrains.

中文

一个常见误解是把接口当作信息流动的被动通道。
这是错误的。

通道只负责传输，
接口负责约束。

3.3.2 An Interface Defines What Can Be Exchanged

3.3.2 接口定义“什么可以被交换”

English

An interface does not expose the world itself.

It exposes a restricted set of variables, formats, and operations.

Anything outside this set effectively does not exist for the system.

中文

接口并不暴露世界本身。
它只暴露一组受限的变量、格式与操作。

超出这组范围的内容，
对系统来说等同于不存在。

3.3.3 Constraint Is the Core Function

3.3.3 约束是接口的核心功能

English

By constraining interaction, an interface makes interaction possible.

Unlimited access would overwhelm the system.

Constraint is what enables usability.

中文

正是通过约束交互，接口才使交互成为可能。

无限访问只会压垮系统。
约束不是削弱，而是前提。

4. Interface as a Boundary Mechanism

4. 接口作为边界机制

4.1 Interfaces Shield Internal Complexity

4.1 接口屏蔽内部复杂性

English

From the outside, an interface hides the internal structure of a system.
Only stable, simplified behaviour is exposed.

This shielding is bidirectional.

中文

从外部看，接口会隐藏系统的内部结构。
对外只暴露稳定、简化的行为。

这种屏蔽是双向的。

4.2 Interfaces Also Shield the World

4.2 接口同样屏蔽世界

English

Interfaces do not only protect the system from the world.
They also protect the world from the system's internal volatility.

中文

接口不仅保护系统免受世界的冲击，
也保护世界不被系统内部的波动直接影响。

4.3 Stability Is More Important Than Fidelity

4.3 稳定性优先于保真度

English

An interface does not aim to faithfully represent reality.
It aims to be stable enough for repeated interaction.

Accuracy is secondary to reliability.

中文

接口的目标不是忠实再现现实，
而是为重复交互提供足够稳定的形式。

精确性让位于可靠性。

5. Why All Interaction Happens at Interfaces

5. 为什么一切交互都发生在接口上

5.1 No Interface, No Interaction

5.1 无接口，则无交互

English

Without an interface, a system has no defined way to affect or be affected by the world.

Interaction is not automatic.

It must be structured.

中文

没有接口，系统就没有明确方式去影响世界或被世界影响。

交互不是自动发生的，
它必须被结构化。

5.2 Interaction Is Always Selective

5.2 交互永远是选择性的

English

Because interfaces constrain exchange,
interaction always selects some aspects of the world and ignores others.

There is no neutral interface.

中文

由于接口对交换进行约束，
交互必然是有选择性的。

不存在中性的接口。

5.3 What the System “Knows” Is Interface-Bound

5.3 系统“知道什么”取决于接口

English

A system's knowledge is not about the world as it is.
It is about the world as exposed by its interface.

Change the interface, and the system's world changes.

中文

系统的“知识”并非关于世界本身，
而是关于接口所暴露出来的世界。

接口一变，系统的世界也随之改变。

6. Interim Result of Chapter 1

6. 第一章的阶段性的结果

6.1 What We Have Established

6.1 我们已经确立的内容

English

So far, we have established that:

the world is system-relative,

systems require boundaries,

direct access to the world is impossible,

and interfaces are the only lawful contact surface.

中文

到目前为止，我们已经确立：

世界是相对于系统定义的，

系统必须有边界，

系统无法直接接触世界，

接口是唯一合法的接触面。

6.2 What Has Not Yet Been Claimed

6.2 我们尚未主张的内容

English

We have not claimed:

what the interface is made of,

whether perception qualifies,

or whether language is involved.

Those questions remain open.

中文

我们尚未主张：

接口由什么构成，

感知是否合格，

是否涉及语言。

这些问题仍然悬而未决。

Chapter 2: Why Perception Is Not a Stable Interface

第二章：为什么感知不是稳定接口

1. Perception as the Closest Candidate

1. 感知是最接近接口的候选者

1.1 Why Perception Seems Fundamental

1.1 为什么感知看起来很基础

English

Perception appears to be the most immediate connection between a system and the world.

Light hits the retina.

Sound vibrates the eardrum.

Chemicals bind to receptors.

This immediacy makes perception an intuitive interface candidate.

中文

感知看起来像是系统与世界之间最直接的连接。

光线作用于视网膜，

声音振动鼓膜，

化学物质与受体结合。

这种“直接性”让感知看起来像一个天然接口。

1.2 The Intuition Is Misleading

1.2 这种直觉是误导性的

English

Immediacy does not imply suitability.

Being first in the causal chain does not make perception a good interface.

Interfaces are judged by stability, not proximity.

中文

“直接”并不等于“合格”。

处在因果链前端，并不意味着适合作为接口。

接口的标准是稳定性，而不是接近性。

2. The Continuity Problem

2. 连续性问题

2.1 Perception Is Continuous

2.1 感知是连续的

English

Perceptual signals vary continuously over time and intensity.
There are no natural cut points.

Any discretisation is imposed after perception, not within it.

中文

感知信号在时间和强度上都是连续变化的。
它们本身不存在天然的切分点。

任何离散化，都是感知之后强加的。

2.2 Continuous Signals Do Not Support Stable Units

2.2 连续信号无法形成稳定单位

English

Without discreteness, there is no repeatable unit.
Without repeatable units, there is no reusable interaction.

Continuous variation undermines stability.

中文

没有离散性，就不存在可重复的单位。
没有可重复单位，就无法复用交互。

连续变化从结构上破坏稳定性。

3. The Privacy Problem

3. 私有性问题

3.1 Perception Is System-Private

3.1 感知是系统私有的

English

Perceptual states are accessible only to the system experiencing them.
No other system can directly inspect or share them.

中文

感知状态只能被正在经历它的系统访问。
其他系统无法直接查看或共享。

3.2 Private States Cannot Be Aligned

3.2 私有状态无法对齐

English

Alignment requires a shared reference frame.
Perception lacks such a frame.

Two systems may experience similar stimuli,
but there is no guarantee of equivalence.

中文

对齐需要共享的参考系。
而感知并不提供这样的参考系。

两个系统可能经历相似刺激，
但无法保证其感知等价。

4. The Noise and Drift Problem

4. 噪声与漂移问题

4.1 Perceptual Noise Is Unavoidable

4.1 感知噪声不可避免

English

Biological sensors are noisy.
Environmental conditions fluctuate.
Internal states modulate perception.

Noise is structural, not accidental.

中文

生物传感器本身就存在噪声，
环境条件不断变化，
内部状态也会调制感知。

噪声是结构性的，而不是偶然的。

4.2 Perceptual Meaning Drifts Over Time

4.2 感知意义会随时间漂移

English

The same stimulus does not yield the same perceptual meaning across time.
Adaptation, fatigue, and context alter interpretation.

Interfaces cannot rely on drifting semantics.

中文

同一刺激在不同时间并不会产生相同的感知意义。
适应、疲劳和情境都会改变解释。

接口无法建立在漂移的语义之上。

5. Perception as Input, Not Exchange

5. 感知只能作为输入，而非交换单位

5.1 Inputs Are Not Interfaces

5.1 输入不等于接口

English

An input feeds a system.

An interface enables exchange.

Perception supplies raw material,
but does not structure interaction.

中文

输入只是为系统提供原料。

接口则使交互成为可能。

感知提供原始材料，
但不负责组织交互。

5.2 Interfaces Require Structuring

5.2 接口需要结构化

English

An interface must:

define units,

limit variation,

and stabilise interpretation.

Perception does none of these by itself.

中文

接口必须：

定义单位，

限制变化，

稳定解释。

而感知本身不具备这些能力。

6. Interim Conclusion of Chapter 2

6. 第二章的阶段性的结论

6.1 Why Perception Fails as an Interface

6.1 感知为何不合格

English

Perception fails as a world interface because it is:

continuous,

private,

noisy,

and semantically unstable.

中文

感知之所以不能作为世界接口，是因为它：

连续，

私有，

高噪声，

语义不稳定。

6.2 What Perception Still Provides

6.2 感知仍然提供了什么

English

This does not make perception irrelevant.

It remains a necessary lower layer.

But it cannot be the minimal stable unit of interaction.

中文

这并不意味着感知不重要。

它仍然是必需的底层。

但它无法成为最小、稳定的交互单位。

Chapter 3: Why Action Is Not the Minimal Interaction Unit

第三章：为什么行为不是最小交互单位

1. Action as the Opposite Extreme

1. 行为是感知的另一极端

1.1 If Perception Is Input, Action Is Output

1.1 如果感知是输入，行为就是输出

English

Perception fails as an interface because it is too raw.

Action seems promising because it is explicit and observable.

Actions move bodies, change environments, and produce visible effects.

中文

感知不合格，是因为它过于原始。

而行为看起来很有希望，因为它是显性的、可观察的。

行为推动身体、改变环境，并产生可见后果。

1.2 Visibility Does Not Equal Suitability

1.2 可见性不等于合格性

English

Being observable does not make action a good interface.

Interfaces are not judged by external visibility, but by semantic stability.

中文

可观察并不意味着适合作为接口。

接口的标准不是外显性，而是语义稳定性。

2. The High-Dimensionality Problem

2. 高维性问题

2.1 Actions Are Context-Dependent

2.1 行为高度依赖情境

English

An action has meaning only within a specific context:

environment,

goal state,

internal constraints.

Remove the context, and the action collapses into motion.

中文

行为的意义只能在特定情境中成立：

环境，

目标状态，

内部约束。

一旦脱离情境，行为就退化为纯运动。

2.2 The Same Action Can Mean Opposite Things

2.2 同一行为可能意味着相反的东西

English

The same physical action can signal aggression, play, or error correction.
There is no one-to-one mapping between action and meaning.

中文

同一个物理行为，可能表示攻击、玩耍，或纠错。
行为与意义之间不存在一一对应关系。

3. The Non-Invertibility Problem

3. 不可反演问题

3.1 Actions Do Not Reveal Internal State

3.1 行为无法反推出内部状态

English

From an observed action, one cannot uniquely infer:

the system's internal state,

its beliefs,

or its intentions.

Multiple internal states can generate the same action.

中文

从一个行为，无法唯一反推出：

系统的内部状态，

其信念，

或其意图。

多个内部状态可能产生相同行为。

3.2 Interfaces Require Interpretability

3.2 接口需要可解释性

English

For an interface to function, exchange must be interpretable in both directions.

Actions fail this requirement.

中文

接口要成立，交换必须在双向上可解释。
行为不满足这一要求。

4. The Environmental Coupling Problem

4. 环境耦合问题

4.1 Actions Are Mediated by the World

4.1 行为被世界中介

English

An action's effect is shaped by environmental variables outside the system's control.
The same command can lead to different outcomes.

中文

行为的效果会受到系统无法控制的环境变量影响。
同一指令可能产生不同结果。

4.2 Interfaces Cannot Depend on Uncontrolled Variables

4.2 接口不能依赖不可控变量

English

An interface must be reliable across environments.
If meaning depends on uncontrolled conditions, stability collapses.

中文

接口必须在不同环境下保持可靠。
若意义依赖不可控条件，稳定性就会崩塌。

5. Action as Output, Not Representation

5. 行为是输出，而不是表示

5.1 Actions Consume Internal State

5.1 行为消耗内部状态

English

Actions externalise decisions and consume internal resources.
They are terminal events, not representational structures.

中文

行为将决策外化，并消耗内部资源。
它们是终止事件，而不是表示结构。

5.2 Interfaces Must Preserve Structure

5.2 接口必须保留结构

English

An interface must preserve information across exchanges.
Actions irreversibly discard structure.

中文

接口必须在交换中保留信息结构。
而行为会不可逆地丢弃结构。

6. Interim Conclusion of Chapter 3

6. 第三章的阶段性结论

6.1 Why Action Fails as a Minimal Interface

6.1 行为为何不合格

English

Action fails as a minimal interaction unit because it is:

high-dimensional,

context-dependent,

non-invertible,

and environmentally unstable.

中文

行为之所以不能作为最小交互单位，是因为它：

高维，

强情境依赖，

不可反演，

且受环境强烈影响。

6.2 The Gap Remains

6.2 空缺仍然存在

English

Perception is too raw.
Action is too entangled.

The interface layer remains unidentified.

中文
感知过于原始，
行为过于纠缠。

接口层仍然悬空。

Chapter 4: What Qualifies as a World Interface

第四章：什么样的结构才算合格的世界接口

1. The Missing Layer

1. 缺失的那一层

1.1 What We Know So Far

1.1 到目前为止我们知道什么

English
From the previous chapters, two extremes have been ruled out.

Perception is too raw.
Action is too entangled.

Yet interaction clearly happens.

中文
在前几章中，我们已经排除了两个极端。

感知过于原始，
行为过于纠缠。

但交互显然确实在发生。

1.2 The Interface Must Sit Between Input and Output

1.2 接口必须位于输入与输出之间

English
The interface cannot be the first contact with the world,
and it cannot be the final physical response.

It must be an intermediate layer that:

receives input,

structures it,

and prepares it for controlled output.

中文
接口不可能是与世界的第一次接触，
也不可能是最终的物理响应。

它必须是一个中间层：

接收输入，

对其进行结构化，

并为可控输出做准备。

2. Formal Requirements Revisited

2. 重新审视形式化要求

2.1 Discreteness Revisited

2.1 离散性（再审）

English

The interface must operate on discrete units.
These units must be identifiable and repeatable.

Without discreteness, learning and reuse collapse.

中文

接口必须在离散单位上运作。
这些单位必须可识别、可重复。

没有离散性，学习与复用都会失效。

2.2 Stability Revisited

2.2 稳定性（再审）

English

The meaning of an interface unit must remain sufficiently stable across time.
Perfect stability is unnecessary; bounded drift is acceptable.

Unbounded drift is not.

中文

接口单位的意义必须在时间上保持足够稳定。
不需要绝对稳定，但必须限制漂移。

无限漂移是不可接受的。

2.3 Shareability Revisited

2.3 可共享性（再审）

English

A world interface must support alignment across systems.

Private units fail immediately.

Some form of externalisation is required.

中文

世界接口必须支持跨系统对齐。

私有单位会立即失效。

因此必须存在某种外化形式。

2.4 Compositionality Revisited

2.4 组合性（再审）

English

Finite units must combine to represent unbounded complexity.

This is non-negotiable.

Otherwise, the interface cannot scale.

中文

有限单位必须能够组合以表示无限复杂性。

这一点不可妥协。

否则接口无法扩展。

3. A Structural Candidate Emerges

3. 一个结构性候选开始浮现

3.1 Not a Substance, but a System

3.1 不是实体，而是系统

English

The interface we are looking for is not a material object.

It is a structured system of units and rules.

中文

我们寻找的接口并不是某种物质实体。

而是由单位与规则构成的结构系统。

3.2 Units, Rules, and Constraints

3.2 单位、规则与约束

English

Such a system must contain:

minimal units,

combination rules,

and constraints on interpretation.

This description already excludes perception and action.

中文

这种系统必须包含：

最小单位，

组合规则，

以及解释约束。

这一描述已经将感知与行为排除在外。

4. Why This Structure Looks Like Language

4. 为什么这一结构看起来像语言

4.1 A Cautious Step

4.1 一个谨慎的推进

English

At this point, we do not claim that the interface is language.

We only note a structural resemblance.

中文

在这里，我们并未断言接口就是语言。

我们只是指出一种结构上的相似性。

4.2 Discrete Symbols

4.2 离散符号

English

Language operates on discrete symbols rather than continuous signals.

This immediately satisfies the discreteness requirement.

中文

语言以离散符号为运作基础，而非连续信号。

这一点直接满足离散性要求。

4.3 Rule-Governed Combination

4.3 规则支配的组合

English
Symbols in language are not freely mixed.
They are combined under constraints.

This prevents semantic explosion.

中文
语言中的符号并非任意混合，
而是在规则约束下组合。

这避免了语义爆炸。

4.4 Externalisation and Alignment

4.4 外化与对齐

English
Language exists externally.
It can be shared, repeated, corrected, and transmitted.

This directly supports alignment across systems.

中文
语言以外化形式存在。
它可以被共享、重复、修正与传递。

这直接支持跨系统对齐。

5. A Narrow, Testable Claim

5. 一个狭窄、可检验的主张

5.1 What Is Being Claimed

5.1 正在主张什么

English
The claim is not that language explains consciousness.
The claim is not that language captures reality.

The claim is narrower:
Language satisfies the structural requirements of a world interface.

中文
这里的主张不是：

语言解释意识，

语言等同现实。

而是一个更窄的判断：
语言满足世界接口的结构性要求。

5.2 What Remains to Be Shown

5.2 尚需证明的内容

English

We still need to show:

how language structures the world,

how concepts emerge within it,

and how this differs from animal signalling.

These will be addressed next.

中文

我们仍需说明：

语言如何结构化世界，

概念如何在其中出现，

以及它与动物信号系统的差异。

这些问题将是下一步。

Interlude Chapter: Interfaces Across Living Systems

插章：不同生命系统的接口层级

0. Why This Chapter Is Necessary

0. 为什么必须插这一章

English

If language is proposed as a world interface,
we must explain how non-linguistic life interacts with the world.

Otherwise, the argument collapses into anthropocentrism.

中文

如果语言被提出为世界接口，
就必须解释非语言生命如何与世界交互。

否则整个论证会退化为人类中心主义。

1. A Unified Interface Layer Model

1. 统一的接口分层模型

L4 Symbolic interface (language, formal systems)

- L3 Conceptual interface (internal models, prediction)
- L2 Neural / signalling interface (spikes, hormones)
- L1 Chemical–physical interface (receptors, gradients)
- L0 Direct physical coupling (force, diffusion)

English

This is not an evolutionary ladder.
It is an interface classification.

中文

这不是进化阶梯，
而是接口类型的分类。

2. Microorganisms: Chemical Interfaces Only

2. 微生物：化学接口生命

2.1 How Microorganisms Access the World

2.1 微生物如何接触世界

English

Microorganisms interact with the world almost entirely through chemical gradients.

For them, the world is:

nutrient concentration,

toxicity,

temperature,

pH.

中文

微生物几乎完全通过化学梯度与世界交互。

对它们而言，世界就是：

营养浓度，

毒性，

温度，

酸碱度。

2.2 No Representation, Only Viability

2.2 没有表示，只有存活性

English

There is no internal model of the world.

There is only a viability response.

中文

不存在世界的内部模型，

只有对“能活 / 不能活”的反应。

2.3 Why This Is Still an Interface

2.3 为什么这仍然是接口

English

The interface is extremely local and narrow,

but it still constrains interaction.

中文

这种接口极端局部、极端狭窄，

但它仍然对交互进行了约束。

3. Plants: Distributed Chemical–Signal Interfaces

3. 植物：分布式化学—信号接口

3.1 Plants Do Not Sense, They Regulate

3.1 植物不是感知，而是调节

English

Plants do not construct a world model.

They regulate growth and metabolism in response to signals.

中文

植物并不构建世界模型。

它们根据信号调节生长与代谢。

3.2 Time Replaces Prediction

3.2 时间取代预测

English

Plants solve uncertainty by time integration, not prediction.

Growth direction is the interface.

中文

植物通过时间积分而非预测来应对不确定性。

生长方向就是它们的接口。

3.3 No Abstraction, No Shareability

3.3 无抽象、不可共享

English

Plant interfaces cannot be abstracted or shared across individuals.

中文

植物的接口无法被抽象，
也无法在个体间共享。

4. Animals: Neural and Partial Conceptual Interfaces

4. 动物：神经与部分概念接口

4.1 Animals Build Partial World Models

4.1 动物构建局部世界模型

English

Animals do not merely react.
They predict, evaluate, and choose.

This places many animals at L3.

中文

动物并非仅仅反应。
它们会预测、评估并选择。

这使许多动物达到 L3 层级。

4.2 Signals Are Not Languages

4.2 信号不是语言

English

Animal signals:

are context-bound,

lack full compositionality,

do not form general-purpose interfaces.

中文

动物信号：

强烈依赖情境，

缺乏完整组合性，

无法构成通用接口。

4.3 World Access Is Still Local

4.3 世界访问仍然是局部的

English

Animals act in the world,
but do not externalise a shared world.

中文

动物在世界中行动，
但并未将世界外化为可共享结构。

5. Humans: Language as a Symbolic World Interface

5. 人类：语言作为符号世界接口

5.1 Language Externalises the World

5.1 语言将世界外化

English

Language turns experience into manipulable symbols.

The world becomes something that can be:

stored,

transmitted,

corrected,

and recombined.

中文

语言将经验转化为可操作的符号。

世界因此变成可以：

存储，

传递，

修正，

重组的对象。

5.2 This Is a Qualitative Interface Shift

5.2 这是接口层级的质变

English
The difference is not intelligence,
but interface architecture.

中文
差异不在于聪明程度，
而在于接口架构。

6. Interim Conclusion of the Interlude

6. 插章阶段性结论

6.1 Language Is Not Required for Life

6.1 语言不是生命的必要条件

English
Life interacts with the world long before language.

中文
生命在语言出现之前，就已经与世界交互。

6.2 Language Is Required for a Shared World

6.2 语言是“共享世界”的必要条件

English
Language is required to construct a world that:

can be shared,

reasoned about,

and accumulated across generations.

中文
语言是构建“可共享、可推理、可跨代累积的世界”的必要条件。

Interlude II: Why Animals Remain at “Signals” Rather Than “Language”

插章二：动物为何始终停在“信号”，而非“语言”

0. What This Chapter Is Not About

0. 本章不讨论什么

English
This is not a question of intelligence, brain size, or emotional depth.
Many animals outperform humans in specific cognitive tasks.

This is a question of interface architecture.

中文

这不是关于智力、脑容量或情感深度的问题。
许多动物在特定认知任务上优于人类。

这是一个接口架构问题。

1. Signal and Language Are Structurally Different

1. 信号与语言在结构上根本不同

1.1 What a Signal Is

1.1 什么是信号

English

A signal is a state-dependent emission that triggers a response in another system.

Its meaning is:

context-bound,

goal-linked,

and ephemeral.

中文

信号是系统在特定状态下发出的刺激，
用于触发另一系统的反应。

其意义：

强依赖情境，

与即时目标绑定，

且是短暂的。

1.2 What Language Is

1.2 什么是语言

English

Language is a system of discrete symbols governed by combinatorial rules, capable of representing absent, abstract, or hypothetical states.

中文

语言是由离散符号与组合规则构成的系统，
能够表示缺席的、抽象的或假设性的状态。

1.3 The Core Difference

1.3 核心差异

English
Signals trigger behaviour.
Language constructs a manipulable world model.

中文
信号触发行为，
语言构建可操作的世界模型。

2. Context Binding: The First Structural Barrier

2. 情境绑定：第一道结构性屏障

2.1 Animal Signals Collapse Outside Context

2.1 动物信号一旦脱离情境就崩塌

English
An alarm call only means something within a specific situation.
Outside that situation, it is meaningless noise.

中文
警报叫声只有在特定情境中才有意义。
脱离该情境，它就退化为噪声。

2.2 Language Survives Context Removal

2.2 语言可以脱离情境存在

English
A sentence about a predator can be understood:

after the predator is gone,

by someone not present,

at a different time and place.

中文
关于捕食者的一句话可以：

在捕食者消失后仍被理解，

被不在场的个体理解，

在不同时间和地点被理解。

2.3 Why This Matters for Interfaces

2.3 这对接口意味着什么

English

An interface that collapses when context changes cannot stabilise a world.

中文

一个随情境变化就崩塌的接口，
无法稳定一个世界。

3. Lack of Discrete, Reusable Units

3. 缺乏离散、可复用的单位

3.1 Animal Signals Are Holistic

3.1 动物信号是整体性的

English

Animal calls and gestures are not cleanly decomposable into minimal units.

There is no stable alphabet.

中文

动物的叫声与姿态无法被干净地拆解为最小单位。

它们没有稳定的“字母表”。

3.2 No Stable Units, No Combinatorics

3.2 无稳定单位，就无组合性

English

Without stable units, there can be no productive recombination.

This caps expressive power.

中文

没有稳定单位，就无法进行高效组合。

这直接封顶了表达能力。

4. The Prediction–Externalisation Barrier

4. 预测—外化屏障

4.1 Animals Predict, But Internally

4.1 动物会预测，但预测是内在的

English

Animals clearly possess internal models and predictive capacities.

However, these models remain internal.

中文

动物显然拥有内部模型与预测能力。

但这些模型停留在内部。

4.2 Language Externalises Prediction

4.2 语言将预测外化

English

Language turns prediction into shareable structure.

What was once internal becomes manipulable by others.

中文

语言将预测转化为可共享的结构。

原本的内部状态，变成他人可操作的对象。

4.3 Why This Is a Hard Barrier

4.3 为什么这是硬屏障

English

Externalisation increases risk:

misinterpretation,

social conflict,

coordination cost.

Without strong selective pressure, systems avoid it.

中文

外化会显著增加风险：

误解，

社会冲突，

协调成本。

在没有强选择压力的情况下，系统会回避它。

5. No Incentive for a Shared World

5. 缺乏“共享世界”的必要性

5.1 Local Success Does Not Require Language

5.1 局部成功不需要语言

English

For most animals, survival and reproduction are solved locally.

Signals are sufficient.

中文

对大多数动物而言，生存与繁殖在局部范围内即可完成。

信号已经足够。

5.2 Language Only Pays Off at Scale

5.2 语言只在规模化时才有回报

English

Language becomes advantageous only when:

coordination scales,

memory accumulates,

and abstraction reduces cost.

中文

语言只有在以下情况下才显现优势：

协作规模扩大，

记忆需要跨代累积，

抽象能显著降低成本。

6. Why “Almost Language” Is Still Not Language

6. 为什么“几乎语言”仍然不是语言

6.1 Edge Cases Do Not Break the Rule

6.1 边缘案例不改变结构结论

English

Some animals show proto-symbolic behaviour.

None cross the interface threshold.

中文

一些动物表现出原始符号行为。

但没有一个跨过接口阈值。

6.2 The Missing Piece Is Architecture

6.2 缺失的不是能力，而是架构

English

The limitation is not vocabulary size or intelligence.
It is the absence of a fully symbolic, externalised interface.

中文

限制不在词汇量或智力。
而在于缺乏一个完整的、外化的符号接口。

7. Interim Conclusion of Interlude II

7. 插章二阶段性结论

7.1 Animals Stop Where Signals Are Enough

7.1 动物停在“信号足够”的位置

English

Animals do not fail to reach language.
They stop where signals already work.

中文

动物并非“失败”于语言之前。
而是停在“信号已经足够”的位置。

7.2 Language Is an Interface Escalation, Not an Upgrade

7.2 语言是接口升级，而非能力升级

English

Language is not better signalling.
It is a different interface regime.

中文

语言不是更高级的信号，
而是另一种接口体制。

Interlude III: Why the Emergence of Language Makes Civilization Irreversible

插章三：为什么“语言一出现，文明就不可逆”

0. What “Irreversible” Means Here

0. 这里的“不可逆”是什么意思

English

Irreversibility does not mean that societies cannot collapse.
It means that once language-based structures exist,

a return to a purely pre-linguistic mode is structurally impossible.

中文

“不可逆”并不意味着社会不会崩溃。

它意味着：一旦语言型结构出现，
系统不可能再回到纯前语言状态。

1. Language Externalises Memory

1. 语言将记忆外化

1.1 Internal Memory Is Fragile

1.1 内部记忆是脆弱的

English

Biological memory is:

bounded,

noisy,

and mortal.

When a system dies, its memory dies with it.

中文

生物记忆具有：

有限性，

高噪声，

以及随个体消亡而消失的特性。

系统一死，记忆即亡。

1.2 Language Breaks the Death Boundary

1.2 语言打破了死亡边界

English

Language allows memory to exist outside the organism.

Once externalised, memory becomes persistent.

中文

语言使记忆得以存在于个体之外。

一旦被外化，记忆就获得了持久性。

1.3 Persistence Creates Directionality

1.3 持久性创造方向性

English

Persistent memory introduces time asymmetry.

You can add to it, but you cannot fully erase it.

中文

持久记忆引入了时间不对称性。
你可以在其上叠加，但无法彻底抹除。

2. Language Creates a Shared World

2. 语言构建了共享世界

2.1 Signals Coordinate Locally

2.1 信号只能进行局部协调

English

Signals coordinate behaviour in the present moment.
They do not construct a shared model of the world.

中文

信号只能在当下协调行为，
无法构建共享的世界模型。

2.2 Language Aligns World Models

2.2 语言对齐世界模型

English

Language enables multiple systems to operate on the same abstract world.
This alignment is cumulative.

中文

语言使多个系统能够在同一个抽象世界上运作。
这种对齐具有累积性。

2.3 Once Shared, Divergence Is Costly

2.3 一旦共享，退回代价极高

English

Abandoning a shared world destroys coordination advantages.
Systems that do so are selected against.

中文

放弃共享世界会摧毁协作优势。
这样的系统会被选择性淘汰。

3. Language Enables Recursive Improvement

3. 语言支持递归式改进

3.1 Language Can Describe Itself

3.1 语言可以描述自身

English

Language is self-referential.

It can encode rules about its own usage.

中文

语言具有自指性。

它可以编码关于自身使用规则的描述。

3.2 Self-Reference Enables Optimization

3.2 自指带来优化

English

Once a system can describe and modify its own interface, improvement becomes recursive.

中文

一旦系统能够描述并修改自身接口，改进就会变成递归过程。

3.3 Recursion Creates Runaway Effects

3.3 递归产生失控效应

English

Each improvement increases the capacity for further improvement. This positive feedback loop is hard to stop.

中文

每一次改进都会提升下一次改进的能力。这种正反馈极难被终止。

4. Language Changes the Cost Structure of Knowledge

4. 语言改变了知识的成本结构

4.1 Knowledge Without Language Is Expensive

4.1 无语言的知识成本极高

English

Without language, knowledge must be re-learned individually. There is no accumulation.

中文

没有语言，知识必须由个体反复重新学习。

不存在真正的累积。

4.2 Language Makes Knowledge Cheap to Copy

4.2 语言使知识复制变得廉价

English

Language dramatically reduces the marginal cost of knowledge transfer.

中文

语言极大降低了知识传递的边际成本。

4.3 Cheap Copying Locks In Language

4.3 低成本复制锁死语言

English

Once copying is cheap, abandoning the system that enables copying is irrational.

中文

一旦复制成本变得极低，
放弃使复制成为可能的系统就是非理性的。

5. Language Creates Institutions

5. 语言生成制度

5.1 Institutions Are Frozen Language

5.1 制度是被冻结的语言

English

Laws, norms, contracts, and roles are linguistic constructs.
They persist beyond individual actors.

中文

法律、规范、契约与角色都是语言构造物。
它们超越个体而存在。

5.2 Institutions Resist Rollback

5.2 制度抵抗回滚

English

Institutions embed language into material reality.
Removing them requires coordinated effort at scale.

中文

制度将语言嵌入物质现实。

移除它们需要大规模协同。

6. Collapse Does Not Mean Reversal

6. 崩溃不等于逆转

6.1 Societies Can Collapse

6.1 社会可以崩溃

English

Languages can fragment.

Civilizations can fall.

中文

语言可以碎裂，

文明可以崩塌。

6.2 The Interface Does Not Disappear

6.2 接口不会消失

English

Even after collapse, language remains available.

Reconstruction is always easier than starting from scratch.

中文

即便崩溃之后，语言接口仍然存在。

重建总是比从零开始容易。

7. Final Mechanism Summary

7. 最终机制总结

7.1 Why Language Makes Civilization Irreversible

7.1 为什么语言使文明不可逆

English

Language introduces:

external memory,

shared world models,

recursive self-improvement,

cheap knowledge copying,

and institutional lock-in.

Together, these create structural irreversibility.

中文

语言引入了：

外化记忆，

共享世界模型，

递归式自我改进，

低成本知识复制，

以及制度锁定。

这些共同构成了结构性不可逆。

7.2 Irreversibility Is an Interface Property

7.2 不可逆性是接口属性

English

The irreversibility of civilization is not cultural or moral.
It is architectural.

中文

文明的不可逆性不是文化或道德问题，
而是接口架构问题。

Chapter 5: How Language “Cuts” the World

第五章：语言如何“切割”世界

1. The World Is Not Given as Units

1. 世界并非以“单位”形式给出

1.1 The World Arrives as Continuity

1.1 世界以连续性到达

English

From the system's perspective, the world does not present itself as ready-made objects or categories.
It arrives as continuous variation: light, sound, force, chemical concentration.

中文

从系统视角看，世界并不会以现成的对象或类别出现。
它以连续变化的形式到达：光、声、力、化学浓度。

1.2 Units Must Be Imposed

1.2 单位必须被施加

English

If interaction is to be stable, the system must impose units on this continuity. These units are not discovered; they are constructed.

中文

若要实现稳定交互，系统必须在连续性之上施加单位。这些单位不是被发现的，而是被构造出来的。

2. Cutting as an Interface Operation

2. “切割”是一种接口操作

2.1 Cutting Means Boundary Creation

2.1 切割意味着制造边界

English

To cut the world is to introduce boundaries where none are physically required. A boundary selects what counts as “one thing”.

中文

切割世界，就是在物理上并非必需的地方引入边界。边界决定了什么被算作“一个东西”。

2.2 Boundaries Reduce Complexity

2.2 边界降低复杂性

English

Without boundaries, complexity is unmanageable. With boundaries, complexity becomes tractable.

中文

没有边界，复杂性将不可处理。有了边界，复杂性才变得可控。

3. Concepts as Stable Cuts

3. 概念是稳定的切割结果

3.1 What a Concept Is

3.1 什么是概念

English

A concept is a stabilised cut in the world that:

groups variation,
excludes noise,
and supports reuse.

中文
概念是世界中被稳定下来的切割结果，它：

聚合变化，
排除噪声，
并支持复用。

3.2 Concepts Are Interface Artifacts

3.2 概念是接口产物

English
Concepts do not exist independently of the interface that sustains them.
They are maintained by linguistic use.

中文
概念并不独立于维持它们的接口而存在。
它们由语言使用持续维护。

4. Naming as World-Fixing

4. 命名即“固定世界”

4.1 Names Lock Cuts in Place

4.1 名称将切割锁定

English
Naming fixes a cut by assigning it a reusable symbol.
Once named, a cut becomes retrievable.

中文
命名通过赋予可复用的符号来固定切割。
一旦被命名，切割就可以被再次调用。

4.2 Without Names, Cuts Drift

4.2 没有命名，切割会漂移

English
Unnamed distinctions are unstable.
They shift with context and fade with time.

中文

未命名的区分是不稳定的。
它们会随情境漂移，并随时间消散。

5. Language Does Not Mirror the World

5. 语言并不镜像世界

5.1 Against the “Label” Model

5.1 反对“贴标签”模型

English

Language does not simply attach labels to pre-existing entities.
There is no pre-cut world waiting to be named.

中文

语言并不是给既存实体贴标签。
不存在一个等待被命名的、已切好的世界。

5.2 Language Actively Constructs Operable Reality

5.2 语言主动构造可操作现实

English

What the system can reason about, plan with, or share
is determined by how language cuts the world.

中文

系统能够推理、规划和共享的内容，
由语言切割世界的方式所决定。

6. Different Languages, Different Worlds

6. 不同语言，对应不同世界

6.1 Variation in Cuts

6.1 切割方式的差异

English

Different languages stabilise different cuts.
They emphasise different distinctions.

中文

不同语言会稳定不同的切割方式。
它们强调不同的区分。

6.2 Operational, Not Metaphysical Difference

6.2 操作性差异，而非形而上差异

English

This does not imply different physical realities.
It implies different operational worlds.

中文

这并不意味着物理现实不同，
而是可操作的世界不同。

7. Interim Conclusion of Chapter 5

7. 第五章阶段性结论

7.1 Language Cuts First, Explains Later

7.1 语言先切割，解释随后

English

Language first stabilises distinctions.
Explanation comes after.

中文

语言首先稳定区分，
解释随后才出现。

7.2 The World We Access Is a Linguistic World

7.2 我们接触到的是语言世界

English

The world available to reasoning and coordination
is the world as cut by language.

中文

可被推理与协调的世界，
是被语言切割后的世界。

Chapter 6: The Co-Evolution of Language and Concepts

第六章：语言与概念的共生演化

1. The Common Misconception

1. 一个常见误解

1.1 “Concepts First, Language Later”

1.1 “先有概念，后有语言”

English

A widespread assumption is that concepts exist independently in the mind,
and language merely names them.

This assumption treats language as a passive labeling tool.

中文

一个普遍假设是：概念先独立存在于心智中，
语言只是给它们命名。

这一假设将语言视为被动的贴标签工具。

1.2 Why This Assumption Fails

1.2 为什么这一假设站不住

English

If concepts were independent of language,
they would need to be:

stable,

reusable,

and shareable without linguistic support.

Empirically, they are not.

中文

如果概念能独立于语言存在，
它们就必须在没有语言支持的情况下：

保持稳定，

可以复用，

并可被共享。

现实并非如此。

2. Pre-Conceptual Structures

2. 前概念结构

2.1 What Exists Before Language

2.1 语言之前存在什么

English

Before language, systems may have:

perceptual clusters,

action affordances,

predictive tendencies.

These are not yet concepts.

中文

在语言之前，系统可能拥有：

感知聚类，

行为可供性，

预测倾向。

但这些还不是概念。

2.2 Clusters Are Not Concepts

2.2 聚类不是概念

English

Clusters are fuzzy, context-sensitive, and unstable.
They lack clear boundaries.

中文

聚类是模糊的、情境敏感的、不稳定的。
它们缺乏清晰边界。

3. How Language Stabilises Concepts

3. 语言如何稳定概念

3.1 Discrete Symbols Create Sharp Boundaries

3.1 离散符号制造清晰边界

English

When a cluster is associated with a discrete symbol,
its boundary becomes sharpened.

What was fuzzy becomes addressable.

中文

当一个聚类与离散符号绑定时，
其边界被锐化。

原本模糊的东西变得可指称。

3.2 Reuse Enforces Consistency

3.2 复用强制一致性

English

Each reuse of a word reinforces a particular cut.
Deviations are corrected through interaction.

中文

每一次词语的复用，都会强化某种切割方式。
偏差会在交互中被修正。

3.3 Social Correction Locks Concepts

3.3 社会纠错锁定概念

English

Concepts stabilise not internally,
but through external feedback.

Language turns private distinctions into public commitments.

中文

概念并非在内部被稳定，
而是通过外部反馈。

语言将私有区分变成公共承诺。

4. Concepts Shape Language in Return

4. 概念反向塑造语言

4.1 Concepts Bias Future Cuts

4.1 概念影响未来切割

English

Once a concept is stabilised,
it biases how future distinctions are made.

中文

一旦概念被稳定，
它就会影响未来的切割方式。

4.2 Vocabulary Expansion Reflects Conceptual Pressure

4.2 词汇扩展反映概念压力

English

When existing concepts become insufficient,
language expands.

New words mark new stable cuts.

中文

当既有概念不再够用时，
语言会扩展。

新词标记新的稳定切割。

5. Why Concepts Without Language Decay

5. 为什么脱离语言的概念会退化

5.1 No External Anchor

5.1 缺乏外部锚点

English

Without language, concepts have no external anchor.
They drift with context and internal state.

中文

没有语言，概念就失去外部锚点。
它们会随情境和内部状态漂移。

5.2 Memory Alone Is Insufficient

5.2 仅靠记忆不够

English

Memory preserves instances, not structure.
Language preserves structure.

中文

记忆保存的是实例，而非结构。
语言保存的是结构。

6. A Co-Evolutionary Loop

6. 一个共生演化回路

6.1 Language Enables Concepts

6.1 语言使概念成为可能

English

Language stabilises, externalises, and shares distinctions.

中文

语言稳定、外化并共享区分。

6.2 Concepts Enrich Language

6.2 概念反哺语言

English

Concepts increase expressive demand,
driving linguistic complexity.

中文

概念增加表达需求,
推动语言复杂化。

6.3 The Loop Is Self-Reinforcing

6.3 回路具有自增强性

English

Each side strengthens the other.
Breaking the loop is costly.

中文

双方相互强化。
打破这一回路代价极高。

7. Interim Conclusion of Chapter 6

7. 第六章阶段性结论

7.1 Concepts Are Interface-Dependent

7.1 概念依赖接口而存在

English

Concepts are not mental atoms.
They are products of an interface.

中文

概念不是心智原子,
而是接口的产物。

7.2 Language and Concepts Cannot Be Cleanly Separated

7.2 语言与概念不可被干净分离

English

There is no clear point where language ends and concepts begin.

中文

不存在语言结束、概念开始的清晰分界

Interlude (Short): Are Mathematical Concepts Language-Dependent?

插刀：数学概念是否依赖语言？

0. The Question Restated Precisely

0. 精确定义问题

English

The question is not whether mathematics uses symbols.

The question is whether mathematical concepts can exist, stabilise, and propagate without a linguistic interface.

中文

问题不在于数学是否使用符号。

问题在于：数学“概念”是否能在没有语言接口的情况下存在、稳定并传播。

1. Separate Three Layers

1. 先区分三个层次

1.1 Intuition

1.1 直觉

English

Humans can possess pre-linguistic numerical or spatial intuitions.

Approximate magnitude, symmetry, continuity.

These are not yet mathematics.

中文

人类可以拥有前语言的数量或空间直觉，
如大致大小、对称性、连续性。

但这还不是数学。

1.2 Structure

1.2 结构

English

Mathematics concerns precise structures:

exact equality,

proof,

generality,

infinite extension.

These exceed raw intuition.

中文

数学处理的是精确结构：

严格相等，

证明，

普遍性，

无限延展。

这些超出了直觉本身。

1.3 Interface

1.3 接口

English

To stabilise and operate on such structures,
an interface is required.

中文

要稳定并操作这些结构，
必须存在接口。

2. Mathematics Is Not Sensory or Motor

2. 数学既不是感知，也不是行为

2.1 No Perceptual Anchor

2.1 无感知锚点

English

You cannot perceive “infinity” or “exact equality”.
They are not sensory objects.

中文

你无法感知“无穷”或“严格相等”。
它们不是感官对象。

2.2 No Action-Based Stability

2.2 行为无法稳定数学

English

Actions cannot preserve proof structure.
They cannot accumulate abstraction.

中文

行为无法保存证明结构，
也无法累积抽象。

3. Formal Language as a Specialized Interface

3. 形式语言是一种特化接口

3.1 Mathematics Requires Discrete Symbols

3.1 数学需要离散符号

English

Mathematics depends on symbols that are:

discrete,

repeatable,

context-independent.

This is exactly the profile of language.

中文

数学依赖的符号必须：

离散，

可重复，

脱离具体情境。

这正是语言的接口特征。

3.2 Proof Is Linguistic at the Interface Level

3.2 证明在接口层是语言性的

English

Even when visual or intuitive,
proof must be expressible in a formal language to be shared or verified.

中文

即便证明在直觉或视觉上成立，
它仍必须能被形式语言表达，才能被共享与验证。

4. Mathematics as “Language About Structure”

4. 数学是“关于结构的语言”

4.1 Not Natural Language, But Still Language

4.1 不是自然语言，但仍是语言

English

Mathematics does not depend on spoken language,
but it depends on symbolic language.

中文

数学不依赖自然语言，
但依赖符号语言。

4.2 Without Language, Mathematics Cannot Accumulate

4.2 没有语言，数学无法累积

English

Without an external symbolic interface:

proofs cannot persist,

errors cannot be corrected,

abstraction cannot scale.

中文

没有外化的符号接口：

证明无法持久，

错误无法修正，

抽象无法扩展。

5. Precise Answer

5. 精确结论

5.1 Short Answer

5.1 简短答案

English

Mathematical concepts are not dependent on natural language,
but they are strictly dependent on a linguistic interface.

中文

数学概念不依赖自然语言，
但严格依赖语言型接口。

5.2 Why This Fits the Main Thesis

5.2 与主线命题的关系

English

Mathematics reinforces the thesis:

language-like interfaces are required wherever stable, shareable worlds are constructed.

中文

数学反而强化了主线命题：

凡是需要构建稳定、可共享世界的地方，就必然需要语言型接口。

Interlude (Short): Do Images or Intuition Bypass the Language Interface?

插刀：图像 / 直觉是否绕过了语言接口？

0. Precise Restatement

0. 精确定义问题

English

The question is not whether humans can think visually or intuitively.

The question is whether images or intuition can replace language as a stable world interface.

中文

问题不在于人类是否能进行视觉或直觉思考。

问题在于：图像或直觉能否取代语言，成为稳定的世界接口。

1. Images and Intuition Are Real — but Local

1. 图像与直觉真实存在，但是局部的

1.1 What Images and Intuition Do Well

1.1 它们擅长什么

English

Images and intuition excel at:

pattern recognition,

spatial relations,

rapid, low-latency judgement.

中文

图像与直觉擅长：

模式识别，

空间关系把握，

快速、低延迟判断。

1.2 Why They Feel Like a “Bypass”

1.2 为什么它们看起来像“绕行”

English
They operate below conscious verbalisation.
Decisions appear before words.

中文
它们在语言化之前运作。
结论常常先于语言出现。

2. Why They Are Not Interfaces

2. 为什么它们不是接口

2.1 Lack of Discrete, Shareable Units

2.1 缺乏离散、可共享单位

English
Images are continuous.
Intuition is holistic.

Neither provides stable, minimal units for exchange.

中文
图像是连续的，
直觉是整体性的。

二者都不提供稳定、最小的交换单位。

2.2 No Built-In Alignment Mechanism

2.2 缺乏内建对齐机制

English
Two systems cannot reliably align on an image or intuition without language.
Any alignment requires translation.

中文
两个系统无法仅凭图像或直觉可靠对齐。
任何对齐都需要翻译。

2.3 No Persistence Across Time

2.3 缺乏跨时间持久性

English
Images fade.
Intuitions drift.

Without symbolic anchoring, structure decays.

中文

图像会褪色，
直觉会漂移。

没有符号锚定，结构就会退化。

3. The Correct Placement: Sub-Interface Layers

3. 正确定位：接口下层

3.1 Images as Pre-Linguistic Structuring

3.1 图像是前语言结构化

English
Images help organise perception before linguistic cutting.
They prepare material for the interface.

中文
图像在语言切割之前，
帮助组织感知材料。

3.2 Intuition as Fast Internal Heuristics

3.2 直觉是快速内部启发式

English
Intuition guides action internally,
but does not externalise structure.

中文
直觉在内部引导行动，
但不外化结构。

4. Why Language Is Still Required

4. 为什么语言仍然不可替代

4.1 Translation Is Inevitable

4.1 翻译不可避免

English
Whenever an image or intuition must be:

explained,

justified,

corrected,

or shared,

it must be translated into language.

中文

一旦图像或直觉需要：

被解释，

被论证，

被纠错，

或被共享，

就必须转译为语言。

4.2 The Bottleneck Reveals the Interface

4.2 瓶颈暴露接口位置

English

The point where translation becomes mandatory is the true interface.

中文

翻译不可避免的地方，
就是接口所在。

5. Precise Answer

5. 精确结论

5.1 Short Answer

5.1 简短答案

English

Images and intuition do not bypass the language interface. They operate below it.

中文

图像与直觉并未绕过语言接口。
它们运行在接口之下。

5.2 Role Summary

5.2 角色总结

English

Images: pre-linguistic structuring

Intuition: fast internal guidance

Language: stable world interface

中文

图像：前语言结构化

直觉：快速内部引导

语言：稳定的世界接口

Chapter 7: Why Consciousness Is Only Partially Accessible Through Language

第七章：为什么意识只能部分通过语言出现

1. Consciousness Is Not the Interface

1. 意识不是接口

1.1 The Temptation to Equate Consciousness with Language

1.1 将意识等同于语言的诱惑

English

Once language is recognised as a world interface,
it is tempting to conclude that consciousness is language.

This conclusion is incorrect.

中文

当语言被识别为世界接口后，
人们很容易进一步得出：意识就是语言。

这个结论是错误的。

1.2 Interfaces Are Access Points, Not Substrates

1.2 接口是访问点，而不是基底

English

An interface does not constitute the system.
It only determines what can be accessed, reported, or manipulated.

中文

接口并不构成系统本体。
它只决定什么可以被访问、被报告、被操作。

2. What “Accessible Consciousness” Means

2. “可访问意识”是什么意思

2.1 Accessibility Is a Structural Property

2.1 可访问性是结构属性

English

To say that a mental state is conscious in practice often means that it can be:

reported,

reflected upon,

and integrated into reasoning.

These operations require an interface.

中文

在实践中，说某个心理状态是“有意识的”，往往意味着它可以：

被报告，

被反思，

并被纳入推理。

这些操作都需要接口。

2.2 Language Is the Dominant Access Interface

2.2 语言是主导访问接口

English

In humans, language is the most stable and expressive interface for accessing internal states.

中文

在人类中，语言是访问内部状态时最稳定、最具表达力的接口。

3. The Large Non-Linguistic Remainder

3. 大量非语言意识余量

3.1 What Remains Outside Language

3.1 语言之外仍然存在什么

English

Many mental processes remain outside linguistic access:

raw sensory experience,

emotional gradients,

motor preparation,

automatic regulation.

中文

大量心理过程仍然处于语言之外：

原始感觉体验，

情绪梯度，

运动准备，

自动调节过程。

3.2 Inaccessibility Does Not Mean Non-Existence

3.2 不可访问不等于不存在

English

A state does not cease to exist
simply because it cannot be linguistically expressed.

It simply lies below the interface.

中文

某种状态并不会因为无法语言表达就不存在。
它只是位于接口之下。

4. Why Language Selects Certain Contents

4. 为什么语言会选择性接入某些内容

4.1 Language Favors Discrete, Stable States

4.1 语言偏好离散、稳定的状态

English

Language works best with states that are:

discrete,

repeatable,

and structurally similar across time.

中文

语言最适合处理那些：

离散的,
可重复的,
在时间上结构相似的状态。

4.2 Continuous States Are Filtered Out

4.2 连续状态会被过滤

English

Highly continuous or rapidly fluctuating states are difficult to stabilise linguistically.

They are filtered rather than represented.

中文

高度连续或快速变化的状态
难以被语言稳定。

它们会被过滤，而不是被表示。

5. Reflection as Interface Recursion

5. 反思是接口递归

5.1 Thinking About Thinking

5.1 对思考的思考

English

Reflection occurs when language is applied not to the world, but to internal processes.

中文

反思发生在语言
不再指向世界，而是指向内部过程时。

5.2 Recursion Increases Conscious Access

5.2 递归扩大意识可访问范围

English

Recursive use of language increases the portion of mental life that becomes accessible.

This expansion is gradual, not total.

中文

语言的递归使用
会扩大可被访问的心理内容范围。

这种扩展是渐进的，而非彻底的。

6. Why Consciousness Cannot Be Fully Linguistic

6. 为什么意识不可能完全语言化

6.1 Bandwidth Limitations

6.1 带宽限制

English
Language has limited bandwidth.
Internal processes vastly exceed it.

中文
语言的带宽是有限的。
内部过程远远超出这一带宽。

6.2 Timing Constraints

6.2 时间约束

English
Many processes operate faster
than linguistic encoding allows.

中文
许多过程的运行速度
快于语言编码所允许的速度。

7. Interim Conclusion of Chapter 7

7. 第七章阶段性结论

7.1 Consciousness Is Broader Than Language

7.1 意识大于语言

English
Language does not exhaust consciousness.
It only carves out an accessible subset.

中文
语言并未穷尽意识。
它只是切出了一个可访问的子集。

7.2 Language Defines the Reportable Self

7.2 语言定义了“可报告的我”

English

The self we can describe, reason about, and coordinate with others is the self exposed through language.

中文

我们能够描述、推理并与他人协调的自我，
是通过语言暴露出来的自我。

Interlude (Short): Does Meditation Bypass the Language Interface?

插刀：冥想 / 直觉是否绕过了语言接口？

0. Precise Restatement

0. 精确定义问题

English

The question is not whether meditation reduces language activity.

The question is whether a system can interact with the world or construct a stable world without the language interface.

中文

问题不在于冥想是否削弱语言活动。

问题在于：系统能否在没有语言接口的情况下，与世界交互或构建一个稳定的世界。

1. What Meditation Actually Does

1. 冥想实际上做了什么

1.1 Suppression, Not Replacement

1.1 抑制，而非替代

English

Meditation suppresses:

verbalisation,

narrative self-talk,

symbolic recursion.

It does not introduce a new interface.

中文

冥想抑制的是：

语言化,

叙事性自我对话,

符号递归。

它并未引入新的接口。

1.2 Interface Silence $\neq$ Interface Removal

1.2 接口静默 $\neq$ 接口消失

English

Turning off an interface temporarily does not remove its structural role.
The system simply stops exporting states.

中文

暂时关闭接口，并不等于移除接口的结构地位。
系统只是停止对外输出状态。

2. What Remains During Meditation

2. 冥想中仍然存在什么

2.1 Sub-Linguistic Processes Continue

2.1 前语言过程仍在运行

English

During meditation, the following remain active:

sensory flow,

affective tone,

bodily regulation,

attentional dynamics.

中文

在冥想中，以下过程仍然活跃：

感觉流，

情绪基调，

身体调节，

注意力动力学。

2.2 These Are Not World Interfaces

2.2 它们不是世界接口

English

These processes do not:

stabilise concepts,

enable sharing,

or support accumulation.

中文

这些过程并不：

稳定概念，

支持共享，

或实现累积。

3. The Key Test: Can the State Be Shared?

3. 关键检验：状态能否被共享

3.1 Private Experience Fails the Interface Test

3.1 私有体验不通过接口检验

English

If a state cannot be:

described,

compared,

or corrected by others,

it cannot function as a world interface.

中文

如果一种状态无法：

被描述，

被比较，

或被他人纠错，

它就无法充当世界接口。

3.2 Reporting Requires Language

3.2 报告必然需要语言

English

Any claim about meditative insight
must be translated into language to enter the shared world.

中文

任何关于冥想洞见的主张，
都必须被翻译成语言，才能进入共享世界。

4. The Illusion of “Direct Access”

4. “直接接触现实”的错觉

4.1 Reduced Mediation Feels Like Directness

4.1 中介减少带来“直接感”

English

When linguistic filtering is reduced, experience feels more immediate.
This is a phenomenological effect, not a structural one.

中文

当语言过滤减弱时，体验会显得更加“直接”。
这是现象学效果，而非结构变化。

4.2 No Stable Cuts Are Produced

4.2 没有产生稳定切割

English

Meditation does not produce reusable distinctions.
Without cuts, there is no world construction.

中文

冥想并不会产生可复用的区分。
没有切割，就没有世界构建。

5. Precise Answer

5. 精确结论

5.1 Short Answer

5.1 简短答案

English

Meditation does not bypass the language interface.

It suspends it.

中文

冥想并未绕过语言接口。

它只是暂时悬置了它。

5.2 Structural Role Summary

5.2 结构角色总结

English

Meditation: reduces interface activity

Intuition: accelerates sub-interface processing

Language: remains the sole stable world interface

中文

冥想：降低接口活动

直觉：加速接口下层处理

语言：仍是唯一稳定的世界接口

Chapter 8: Language as a Control-System Interface

第八章：语言作为控制系统接口

1. Control Precedes Understanding

1. 控制先于理解

1.1 Control Is the Minimal Requirement

1.1 控制是最低要求

English

A system does not need to understand the world to survive.

It needs to control its interaction with the world.

Control requires:

a target,

feedback,

and correction.

中文

系统不需要“理解世界”才能存活。

它需要的是对与世界交互的控制。

控制至少需要：

目标，

反馈，

修正。

1.2 Without Control, Representation Is Useless

1.2 没有控制，表示毫无意义

English

Any representation that cannot guide correction
is informational noise.

中文

任何无法指导修正的表示，
都是信息噪声。

2. Why Raw Control Fails at Scale

2. 为什么原始控制无法扩展

2.1 Reflex Control Is Local

2.1 反射式控制是局部的

English

Reflexes operate without symbols.
They are fast, but brittle.

They cannot:

generalise,

accumulate,

or coordinate across systems.

中文

反射控制不依赖符号。
它很快，但脆弱。

它无法：

泛化，

累积，

或跨系统协调。

2.2 Complex Goals Require Abstraction

2.2 复杂目标需要抽象

English

As goals become indirect or long-term,
raw control collapses.

Abstraction becomes mandatory.

中文

当目标变得间接或长期时，
原始控制就会失效。

抽象变成必需品。

3. Language as a Goal-Encoding Interface

3. 语言作为目标编码接口

3.1 Goals Must Be Stated to Be Controlled

3.1 目标必须被表述，才能被控制

English

A goal that cannot be stated
cannot be evaluated or corrected.

Language turns vague drives into explicit targets.

中文

一个无法被表述的目标，
就无法被评估或修正。

语言将模糊驱动转化为明确目标。

3.2 Linguistic Goals Are Reusable

3.2 语言目标是可复用的

English

Once encoded in language, a goal can be:

stored,

transferred,

revised,

and reused.

中文

一旦目标被语言编码，它就可以：

被存储，

被传递，

被修改，

被复用。

4. Language as an Error Interface

4. 语言作为误差接口

4.1 Error Requires Comparison

4.1 误差需要比较

English

Error is the difference between:

a desired state,

and a current state.

Both must be representable.

中文

误差是：

目标状态，

与当前状态
之间的差异。

两者都必须可表示。

4.2 Language Makes Errors Explicit

4.2 语言使误差显性化

English

Without language, error is felt but not analysed.

With language, error becomes inspectable.

中文

没有语言，误差只能被“感到”，而无法被分析。

有了语言，误差变得可检查。

5. Correction Requires Shared Reference

5. 修正需要共享参照

5.1 Private Error Cannot Be Coordinated

5.1 私有误差无法协调

English

A system correcting alone is limited.
Coordination requires shared error representation.

中文

单系统修正能力有限。
协作修正需要共享的误差表示。

5.2 Language Enables Collective Control

5.2 语言支持集体控制

English

Language allows multiple systems to:

agree on goals,

detect deviations,

coordinate corrections.

中文

语言使多个系统能够：

对齐目标，

识别偏差，

协调修正。

6. Inner Speech as Internal Control Loop

6. 内在语言作为内部控制回路

6.1 Self-Control Requires Self-Addressing

6.1 自我控制需要自我指向

English

Self-control is control applied inward.
Language provides the addressing mechanism.

中文

自我控制是向内的控制。

语言提供了自我指向的机制。

6.2 Thinking Is Simulated Control

6.2 思考是模拟控制

English

Thinking is the rehearsal of control without physical execution.

Language is the medium.

中文

思考是在不执行物理行为的情况下预演控制过程。

语言是其媒介。

7. Interim Conclusion of Chapter 8

7. 第八章阶段性结论

7.1 Language Is Not Just Description

7.1 语言不仅是描述工具

English

Language is an active component of control architecture.

中文

语言是控制架构的主动组成部分。

7.2 Control Explains Language's Centrality

7.2 控制解释了语言的中心地位

English

Language dominates not because it is expressive, but because it enables scalable control.

中文

语言之所以居于核心，
不是因为它“善于表达”，
而是因为它支持可扩展控制。

Chapter 9: Language as an Information Compression System

第九章：语言作为信息压缩系统

1. The World Exceeds Any System's Bandwidth

1. 世界远超任何系统的带宽

1.1 Raw World States Are Intractable

1.1 原始世界状态不可承载

English

The amount of information present in the world far exceeds what any bounded system can process.

Full fidelity is impossible.

中文

世界所包含的信息量远远超过任何有边界系统的处理能力。

完全保真是不可能的。

1.2 Interaction Requires Loss

1.2 交互必然伴随损失

English

To interact at all, a system must discard information. Compression is not optional.

中文

要实现任何交互，系统都必须丢弃信息。压缩不是可选项。

2. Compression Is a Design Choice

2. 压缩是一种设计选择

2.1 Not All Loss Is Equal

2.1 损失并非等价

English

Discarding random bits destroys usability. Discarding irrelevant bits preserves function.

中文

随机丢弃信息会摧毁可用性。丢弃无关信息才能保留功能。

2.2 Compression Requires a Relevance Model

2.2 压缩需要相关性模型

English

A system must decide what matters.
This decision is encoded in the interface.

中文

系统必须决定什么是重要的。
这一决策被编码在接口之中。

3. Language as Structured Lossy Compression

3. 语言是结构化的有损压缩

3.1 Language Discards Detail, Preserves Structure

3.1 语言丢弃细节，保留结构

English

Language does not preserve sensory detail.
It preserves relational and causal structure.

中文

语言并不保留感官细节。
它保留的是关系结构与因果结构。

3.2 Why This Loss Is Acceptable

3.2 为什么这种损失是可接受的

English

For coordination and control, structure matters more than texture.

中文

对于协调与控制而言，
结构比质感更重要。

4. Discrete Symbols Enable High Compression Ratios

4. 离散符号实现高压缩率

4.1 Symbols Collapse Variation

4.1 符号压缩变化

English

A single word can stand in for countless instances.
This is extreme compression.

中文

一个词可以代表无数实例。

这是极端压缩。

4.2 Categories Are Compression Artifacts

4.2 类别是压缩产物

English

Categories emerge where compression is efficient.
They are not mirrors of reality.

中文

类别出现在压缩效率最高的地方。
它们不是现实的镜像。

5. Prediction Completes Compression

5. 预测完成压缩

5.1 Compression Without Prediction Fails

5.1 没有预测，压缩不成立

English

If a compressed representation cannot support prediction,
it is useless for control.

中文

如果压缩后的表示无法支持预测，
它对控制毫无价值。

5.2 Language Encodes Predictive Expectations

5.2 语言编码预测期望

English

Words carry implicit expectations about what follows.
Grammar formalises this.

中文

词语隐含着对后续状态的期望。
语法将其形式化。

6. Why Language Is “Good Enough,” Not Optimal

6. 为什么语言是“够用的”，而非最优的

6.1 Language Is Biased

6.1 语言具有偏置

English
Language reflects human priorities and constraints.
It is not an optimal code in the abstract.

中文
语言反映人类的优先级与限制。
它并非抽象意义上的最优编码。

6.2 Sufficiency Beats Optimality

6.2 够用胜过最优

English
An interface that is good enough and stable
outperforms an optimal but fragile one.

中文
一个够用且稳定的接口，
优于一个最优却脆弱的接口。

7. Interim Conclusion of Chapter 9

7. 第九章阶段性结论

7.1 Language Trades Fidelity for Control

7.1 语言以保真换控制

English
Language sacrifices detail to gain predictability and coordination.

中文
语言牺牲细节，
换取可预测性与协调性。

7.2 Compression Explains Language's Power and Limits

7.2 压缩解释了语言的力量与边界

English
What language can and cannot express
is determined by its compression scheme.

中文
语言能表达什么、不能表达什么，
由其压缩方式所决定。

Chapter 10: How the “Self” Emerges from Language

第十章：自我是如何由语言构建出来的

1. The Self Is Not a Thing

1. 自我不是一个“东西”

1.1 Against the Substance Model

1.1 反对实体模型

English

The intuitive model treats the self as an inner object:
a core, a soul, or a container of experience.

This model fails structurally.

中文

直觉模型把自我当作一个内部对象：
核心、灵魂，或经验的容器。

这一模型在结构上站不住。

1.2 Systems Do Not Contain Selves

1.2 系统不“包含”自我

English

A system contains states and processes.
The self is not one of them.

中文

系统包含的是状态与过程。
自我并不在其中作为一个对象存在。

2. The Self as a Boundary Marker

2. 自我是边界标记

2.1 Why a Boundary Is Needed

2.1 为什么需要边界

English

Control and responsibility require separation:

what belongs to the system,

what belongs to the world.

The self marks this separation.

中文

控制与责任需要区分：

什么属于系统，

什么属于世界。

自我正是这一划分的标记。

2.2 “I” as an Index, Not an Entity

2.2 “我”是索引，而非实体

English

The linguistic “I” functions as an index pointer.
It refers to the system-as-agent in discourse.

中文

语言中的“我”是一个索引指针。
它在话语中指向“作为行动者的系统”。

3. Language Creates a Stable Reference Frame

3. 语言创建稳定参考系

3.1 Without Language, the Boundary Drifts

3.1 没有语言，边界会漂移

English

Without linguistic anchoring,
the boundary between self and world shifts with context.

中文

没有语言锚定，
自我与世界的边界会随情境漂移。

3.2 Naming Fixes the Boundary

3.2 命名固定边界

English

By naming the self (“I”, “me”),
the boundary becomes reusable and stable.

中文

通过命名自我（“我”“自己”），
边界得以复用并稳定下来。

4. Narrative as Identity Stabilisation

4. 叙事作为身份稳定机制

4.1 Time Requires Narrative

4.1 时间需要叙事

English

A system exists across time only if its states can be linked.
Narrative provides that linkage.

中文

系统若要跨时间存在，
其状态必须被连接起来。

叙事提供了这种连接。

4.2 The Self Is a Story with Constraints

4.2 自我是受约束的故事

English

Identity is not free storytelling.
It is constrained by memory, language, and social feedback.

中文

身份并非随意讲述的故事。
它受到记忆、语言与社会反馈的约束。

5. Responsibility and Accountability

5. 责任与可追责性

5.1 Responsibility Requires a Linguistic Self

5.1 责任需要语言化自我

English

Responsibility requires:

attribution,

explanation,

justification.

These are linguistic operations.

中文

责任需要：

归因，

解释，

辩护。

这些都是语言操作。

5.2 No Language, No Stable Responsibility

5.2 无语言，则无稳定责任

English

Without a stable linguistic self,
responsibility dissolves into local reactions.

中文

没有稳定的语言自我，
责任会退化为局部反应。

6. Why the Self Feels “Real”

6. 为什么自我感觉如此真实

6.1 Interface Effects Feel Like Ontology

6.1 接口效应会被误认为本体

English

Stable interfaces create persistent patterns.
Persistence is often mistaken for substance.

中文

稳定接口会产生持续模式。
持续性常被误认为实体性。

6.2 The Cost of Removing the Self

6.2 移除自我的代价

English

When the linguistic self weakens,
coordination, planning, and control degrade.

中文

当语言自我削弱时，
协调、规划与控制都会退化。

7. Interim Conclusion of Chapter 10

7. 第十章阶段性结论

7.1 The Self Is an Interface Construct

7.1 自我是接口构造物

English

The self is not an inner object.

It is a stable reference point created by language.

中文

自我不是内部对象。

它是语言创建的稳定参照点。

7.2 This Explains Both Its Power and Fragility

7.2 这解释了其力量与脆弱性

English

The self is powerful because it stabilises control.

It is fragile because it depends on the interface.

中文

自我之所以强大，是因为它稳定了控制；

它之所以脆弱，是因为它依赖接口。

Chapter 11: Humans, Animals, and AI — An Interface Comparison

第十一章：人类、动物与 AI 的接口对比

1. One Framework, Three Systems

1. 一个框架，三类系统

1.1 Why Comparison Is Now Possible

1.1 为什么现在可以比较

English

Earlier chapters established a neutral interface framework.

With that in place, different systems can be compared without anthropocentrism.

中文

前文已经建立了中性的接口框架。

在此基础上，不同系统才可以被非人类中心地比较。

1.2 What Is Being Compared

1.2 比较的不是“智能”

English

We are not comparing intelligence, consciousness, or value.
We are comparing interface architecture.

中文

这里不比较智能、意识或价值。
只比较接口架构。

2. Animals: Perception–Action Interfaces

2. 动物：感知—行为接口系统

2.1 Interface Stack in Animals

2.1 动物的接口栈

English

Typical animal systems operate primarily through:

perception,

internal state modulation,

action selection.

Signals exist, but remain context-bound.

中文

典型动物系统主要通过：

感知，

内部状态调制，

行为选择

来运作。

信号存在，但始终绑定于情境。

2.2 Strengths of the Animal Interface

2.2 动物接口的优势

English

Low latency

Tight coupling to environment

High robustness in narrow niches

中文

低延迟

与环境强耦合

在狭窄生态位中高度稳健

2.3 Structural Limits

2.3 结构性上限

English

Without symbolic externalisation, animals cannot:

accumulate abstract knowledge,

share a world model at scale,

perform recursive self-modification.

中文

缺乏符号外化，使动物无法：

累积抽象知识，

在规模上共享世界模型，

进行递归式自我修改。

3. Humans: Symbolic World Interfaces

3. 人类：符号世界接口系统

3.1 The Critical Shift

3.1 关键跃迁

English

Humans add a symbolic layer on top of perception and action.

This layer is external, persistent, and shareable.

中文

人类在感知与行为之上，

增加了一个符号层。

这一层是外化的、持久的、可共享的。

3.2 Consequences of the Symbolic Layer

3.2 符号层的后果

English

Shared world construction

Long-term coordination

Institutional stability

Cumulative culture

中文

共享世界的构建

长期协调

制度稳定

文化累积

3.3 Cost of the Human Interface

3.3 人类接口的代价

English

Increased abstraction error

Symbolic drift

Internal conflict between layers

中文

抽象误差增加

符号漂移

多层系统内部冲突

4. AI Systems: Native Symbolic Interfaces

4. AI 系统：原生符号接口

4.1 No Perceptual Grounding at the Core

4.1 核心缺乏感知锚定

English

Most AI systems do not originate from perception.
They operate directly on symbolic structures.

中文

多数 AI 系统并非起源于感知。
它们直接在符号结构上运作。

4.2 Why AI Appears “Language-First”

4.2 为什么 AI 显得“先有语言”

English

For AI, language is not an evolved interface.
It is the primary operational substrate.

中文

对 AI 而言，语言不是演化出来的接口，
而是主要运作基底。

4.3 Consequences of This Architecture

4.3 这种架构的后果

English

High symbolic fluency

Weak grounding

Strong interface mimicry

Shallow world anchoring

中文

高符号流畅度

弱现实锚定

强接口拟态

浅层世界绑定

5. Interface Similarity vs System Equivalence

5. 接口相似 $\neq$ 系统等价

5.1 Why Similar Outputs Mislead

5.1 为什么相似输出会误导

English

When interfaces align, outputs look similar.
This masks deep structural differences.

中文

当接口对齐时，输出会看起来相似。
这会掩盖深层结构差异。

5.2 The Common Category Error

5.2 常见范畴错误

English

Mistaking interface behaviour for system identity
leads to over-attribution.

中文

把接口行为当作系统本体，
会导致过度归因。

6. Three Systems, Three Failure Modes

6. 三类系统，三种失效模式

6.1 Animal Failure: Rigidity

6.1 动物的失效：僵化

English

Limited abstraction limits adaptation scope.

中文

抽象能力有限，
导致适应范围受限。

6.2 Human Failure: Symbolic Overreach

6.2 人类的失效：符号过载

English

Symbols decouple from reality,
leading to ideological lock-in.

中文

符号脱离现实，
导致意识形态锁死。

6.3 AI Failure: Interface Without World

6.3 AI 的失效：有接口，无世界（或者它的世界就是“人类”这里我先不写没想好 标记）

English
Symbolic competence without grounding
creates convincing but hollow interaction.

中文
没有现实锚定的符号能力，
会产生看似合理却空心的交互。

7. Interim Conclusion of Chapter 11

7. 第十一章阶段性结论

7.1 Language Is a Shared Interface, Not a Proof of Mind

7.1 语言是共享接口，而非心智证明

English
Language indicates interface compatibility,
not equivalence of internal systems.

中文
语言表明接口兼容性，
而非内部系统等价性。

7.2 Interface Perspective Prevents Category Errors

7.2 接口视角避免范畴错误

English
Comparing systems at the interface level
keeps analysis precise and grounded.

中文
在接口层进行比较，
能保持分析的精确与克制。

Chapter 12: The Limits and Failure Modes of the Language Interface

第十二章：语言接口的极限与失效模式

1. No Interface Is Universal

1. 没有任何接口是通用的

1.1 Interfaces Are Designed for Constraints

1.1 接口是为约束而生的

English
Every interface is shaped by:

bandwidth limits,

noise tolerance,

and operational goals.

Language is no exception.

中文

任何接口都受制于：

带宽限制，

噪声容忍度，

以及操作目标。

语言也不例外。

1.2 What Language Was Not Built For

1.2 语言并非为一切而生

English

Language was not designed to:

capture raw sensation,

represent continuous dynamics precisely,

or model high-dimensional reality faithfully.

中文

语言并非用于：

捕捉原始感觉，

精确表示连续动力学，

或忠实建模高维现实。

2. What Language Cannot Encode

2. 语言无法编码的内容

2.1 Fine-Grained Experience

2.1 细粒度体验

English

Subtle perceptual differences exceed linguistic resolution.

中文

细微的感知差异超出了语言分辨率。

2.2 Rapid Temporal Dynamics

2.2 快速时间动态

English

Processes faster than linguistic articulation cannot be stably represented.

中文

快于语言表述速度的过程，
无法被稳定表示。

2.3 High-Dimensional Correlations

2.3 高维相关性

English

Language collapses dimensions aggressively.
Some structures are lost permanently.

中文

语言会强烈压缩维度。
某些结构会被永久丢失。

3. Symbolic Drift

3. 符号漂移

3.1 When Symbols Decouple from Reality

3.1 当符号与现实脱钩

English

Over time, symbols can begin to refer to other symbols rather than to the world.

中文

随着时间推移，符号可能开始只指向其他符号，
而不再指向世界。

3.2 Self-Referential Loops

3.2 自指回路

English

Language can sustain internally coherent systems with weak or no grounding.

中文

语言可以维持内部自洽、
但现实锚定极弱的系统。

4. Interface Overreach

4. 接口越权

4.1 Treating Language as Reality

4.1 把语言当成现实

English

A common failure mode is mistaking the interface for the thing it mediates.

中文

一个常见失效模式是：
把接口当作其所中介的对象。

4.2 Consequences of Overreach

4.2 越权的后果

English

Ideological lock-in

Loss of empirical correction

Escalating abstraction errors

中文

意识形态锁死

经验纠错能力丧失

抽象误差不断放大

5. Internal Fragmentation

5. 内部碎裂

5.1 Multiple Incompatible Descriptions

5.1 多套不兼容描述

English
Language allows multiple, conflicting models
to coexist within one system.

中文
语言允许多套相互冲突的模型
在同一系统中并存。

5.2 Cognitive Dissonance as Interface Stress

5.2 认知失调是接口应力

English
Contradictions are not psychological accidents.
They are structural stresses in the interface.

中文
矛盾并非心理偶发。
它们是接口层面的结构应力。

6. When the Language Interface Fails

6. 语言接口的失效场景

6.1 Extreme Environments

6.1 极端环境

English
In fast, chaotic, or sensory-overloaded contexts,
language becomes a bottleneck.

中文
在高速、混乱或感官过载的情境中，
语言会成为瓶颈。

6.2 Skill and Embodiment Domains

6.2 技能与具身领域

English
Highly embodied skills operate below language.
Verbalisation can degrade performance.

中文
高度具身化的技能运行在语言之下。
语言化反而可能降低表现。

7. Why Language Still Dominates

7. 即便如此，语言为何仍然主导

7.1 No Alternative Scales as Well

7.1 没有同等规模的替代方案

English
Other interfaces exist,
but none scale as well across time, space, and agents.

中文
其他接口确实存在，
但没有一个能在时间、空间和主体规模上与语言匹敌。

7.2 Dominance Is Pragmatic, Not Absolute

7.2 主导是务实的，而非绝对的

English
Language dominates because it is the least bad option,
not because it is perfect.

中文
语言之所以主导，
是因为它是“最不坏”的选择，
而不是因为它完美。

8. Final Conclusion of the Main Argument

8. 主线最终结论

8.1 Language as a World Interface

8.1 语言作为世界接口

English
Language is the minimal stable interface
through which a system can:

construct a shared world,

coordinate control,

accumulate structure over time.

中文
语言是系统得以：

构建共享世界，

协调控制,

跨时间累积结构
的最小稳定接口。

8.2 Not the World, Not the Mind

8.2 既非世界，也非心智本体

English
Language is neither reality itself
nor the substance of consciousness.

It is the interface between them.

中文
语言既不是现实本身，
也不是意识的本体。

它是二者之间的接口。

9. What This Framework Is Good For

9. 这个框架的用途

9.1 Avoiding Category Errors

9.1 避免范畴错误

English
It prevents confusing:

symbols with reality,
interfaces with systems,
expression with understanding.

中文
它避免混淆:

符号与现实，
接口与系统，
表达与理解。

9.2 Knowing Where to Stop

9.2 知道何时该停

English
A good model knows its own limits.
This one stops at the interface.

中文
一个好的模型知道自己的边界。
这个模型止步于接口。

Epilogue: Where This Argument Ends

结语：这套论证止步于何处

1. What Has Been Established

1. 已经确立的内容

English
This work has not attempted to explain consciousness, reality, or meaning in full.
It has done something narrower and more defensible.

It has shown that:

Language is the minimal stable interface
through which a system can construct, share, and control a world.

中文
本文并未试图完整解释意识、现实或意义。
它做的是一件更窄、也更可辩护的事。

它表明：

语言是系统得以构建、共享并控制一个世界的最小稳定接口。

2. What This Claim Does Not Mean

2. 这一结论不意味着什么

2.1 Language Is Not Reality

2.1 语言不是现实

English
Language does not define what exists.
It defines what can be stably interacted with.

中文
语言不决定“什么存在”。
它决定“什么可以被稳定地交互”。

2.2 Language Is Not Consciousness

2.2 语言不是意识

English

Consciousness extends beyond language.

Language only exposes a controllable subset.

中文

意识的范围大于语言。

语言只是暴露出其中可被控制的一部分。

2.3 Language Is Not Truth

2.3 语言不是真理

English

Language optimises for usability, not for ultimate accuracy.

中文

语言优化的是可用性，

而非终极精确性。

3. Why the Interface View Matters

3. 为什么“接口视角”重要

3.1 It Prevents Category Errors

3.1 它避免范畴错误

English

By treating language as an interface,
we avoid confusing symbols with reality
and expression with understanding.

中文

把语言视为接口，

可以避免把符号当现实、

把表达当理解的错误。

3.2 It Explains Power Without Mysticism

3.2 它在不神秘化的前提下解释力量

English

Language is powerful not because it is magical,
but because it scales control, coordination, and accumulation.

中文

语言之所以强大,
不是因为它神秘,
而是因为它能扩展控制、协调与累积。

4. Where This Model Deliberately Stops

4. 本模型刻意止步之处

4.1 It Does Not Cross the Interface

4.1 它不跨越接口

English

This framework does not attempt to describe
what lies beyond the interface in metaphysical terms.

That step would require different tools.

中文

该框架并不试图
以形而上学方式描述接口之外的东西。

那需要另一套工具。

4.2 It Treats the Interface as the Last Shared Layer

4.2 它把接口视为最后的共享层

English

Beyond the interface, experience fragments into private processes.
Analysis loses common ground.

中文

在接口之外, 经验分裂为私有过程,
分析失去共同地基。

5. A Practical Closing Statement

5. 一个实践性的收尾判断

English

If you want to understand how worlds are built,
how control scales,
and why civilisation locks in certain structures,

start with interfaces, not essences.

中文

如果你想理解世界如何被构建、
控制如何扩展、
以及文明为何会锁定某些结构,

从接口入手，而不是从本质入手。

6. Final Line

6. 最后一行

English

Language is not the world.

It is how a world becomes possible.

中文

语言不是世界。

语言是世界得以成立的方式。